Physics Worksheet

Gravity

Name: _____ **Student #:** _____ **Student Email #:** _____

1. Two small objects with different masses, m_1 and m_2, are located a distance r from a nearby star. What can you say about the magnitude of the gravitational field and the magnitude of the gravitational force on m_1 and m_2?

2. Honza has decided to embark on a great quest: climbing Mount Everest. Out of scientific interest, he decides to take a spring-scale with him and weigh himself at sea level and at the top of the mountain. What would his scale read for his weight at sea level and at the top of Mount Everest (up to 4 decimal points)? His true weight is 80 kg. Assume the spring scale is coded for a g value of $9.8\,\mathrm{m\,s^{-2}}$.

3. One of Jupiter's moons, Io, has a semi-major axis of $2.81 \times 10^{-}3AU$. What is the period of its orbit?